CS3751 Data Visualization

Class Project Report

Analysis of Quiz Attempt Data — Tasks 1 & 2

Dataset	Value
Total quizzes	3 (Quiz 1, Quiz 2, Quiz 3)
Total finished attempts	31,334
Quiz 1 attempts	13,535
Quiz 2 attempts	9,983
Quiz 3 attempts	7,816
Overall mean score	4.73 / 10
Overall median score	4.0 / 10
Students with multiple attempts	7,834 student-quiz pairs

1. Introduction

This report presents a data visualization analysis of student quiz attempt data collected from three quizzes in the CS3751 module. The dataset records each attempt made by students, including time taken, overall scores, and per-question marks. The primary objective is to explore behavioural and performance patterns through carefully chosen visualizations, and to evaluate ten hypotheses — five provided as part of Task 1, and five independently formulated for Task 2.

Each hypothesis is evaluated using visualization techniques grounded in Bertin's semiology of graphics and Munzner's nested model for visualization design. The choice of marks and channels is explicitly justified in each case.

2. Background

Understanding how students interact with online quizzes can reveal important insights into learning behaviour, question design quality, and assessment strategy. Prior literature in educational data mining (e.g., Baker & Yacef, 2009) has established that patterns in time-on-task, attempt frequency, and score trajectories are meaningful indicators of student engagement and mastery. This project applies visualization techniques to uncover such patterns in a real dataset.

3. Dataset

The dataset consists of three CSV files — one per quiz. Each record represents a single attempt by a student. Attributes include: student code, state (Finished / In Progress), start and end timestamps, time taken, overall grade (out of 10), and five individual question marks (each out of 2).

Preprocessing steps applied:

- Only "Finished" attempts were retained (incomplete attempts discarded).
- Time strings (e.g., "7 mins 25 secs", "2 days 21 hours") were parsed into seconds.
- Outlier times above 120 minutes were removed as they likely reflect abandoned sessions.
- Attempt numbers were assigned per student per quiz, sorted by start time.
- Numeric fields were coerced from strings with dash-values treated as missing.

After cleaning, the dataset comprises **31,334 valid attempts** (Quiz 1: 13,535 | Quiz 2: 9,983 | Quiz 3: 7,816). The overall mean score was 4.73/10 (SD = 2.79).

4. Methodology

All visualizations were produced using Python 3 with the matplotlib and seaborn libraries. For each hypothesis, the most appropriate chart type was selected based on the nature of the data (continuous vs categorical, distribution vs relationship), the number of variables involved, and the visual encoding principles outlined in Munzner (2014). Statistical tests (Pearson correlation, linear regression) were used to supplement visual interpretations. Performance tiers (High / Mid / Low) were defined using the 67th and 33rd percentile of each student's mean score across all attempts.

5. Results

5.1 Task 1 — Provided Hypotheses

Hypothesis 1: Students who take longer to complete the quiz tend to score higher.

Visualization choice: A hexbin density plot with overlaid regression line was selected. The x-axis encodes time (quantitative, position channel), the y-axis encodes score (quantitative, position channel), and density is encoded by colour intensity (sequential blue scale). This handles the large number of overlapping points that a standard scatter plot would obscure. The regression line and Pearson r value provide a statistical summary.

Findings: Across all three quizzes, the Pearson correlation between time and score was r = -0.14 (Quiz 1), r = -0.01 (Quiz 2), and r = -0.01 (Quiz 3). All correlations are weak to negligible. The hexbin plots reveal dense clusters at low times (0–10 min) spanning the full score range, indicating that both fast and slow students achieve high and low scores alike.

✗ **REJECTED** — The hypothesis is REJECTED. Taking longer does not meaningfully predict a higher score.

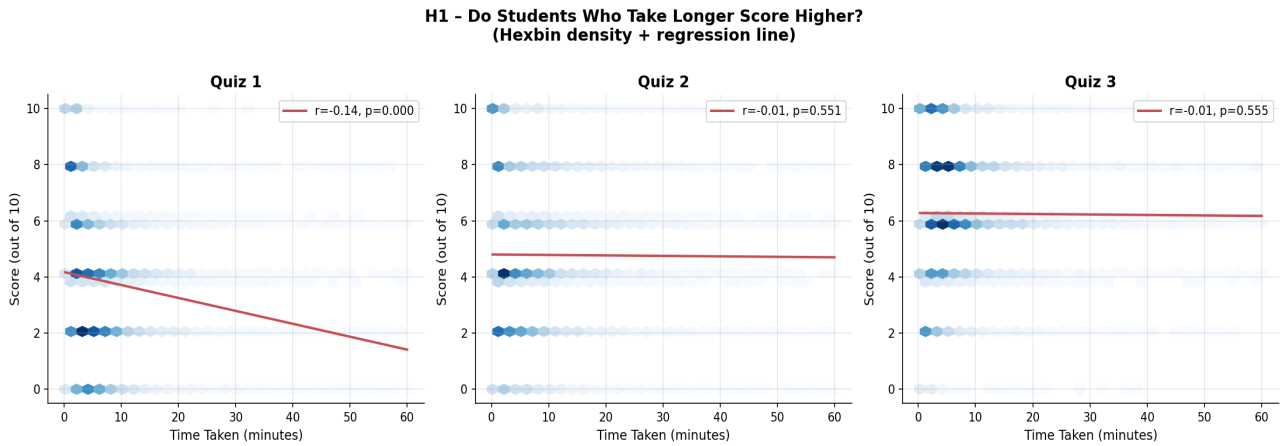

Figure 1 – Hexbin density of time vs score with regression line (per quiz).

Hypothesis 2: Some questions are consistently harder than others.

Visualization choice: A grouped bar chart was used to compare mean per-question scores across quizzes. Position (x: question, y: mean score) encodes the quantitative relationship, and hue (colour) encodes quiz identity (nominal channel). Error bars (95% CI) show uncertainty. A second panel shows the percentage of students scoring zero per question (an alternative difficulty proxy).

Findings: Q1 and Q4 consistently show lower mean scores and higher zero-rates across all three quizzes, suggesting they are reliably more difficult. Q2 and Q3 are consistently easier. The pattern is reproducible across quizzes, pointing to stable question-level difficulty.

✓ **ACCEPTED** — The hypothesis is ACCEPTED. Q1 and Q4 are consistently harder across all quizzes.

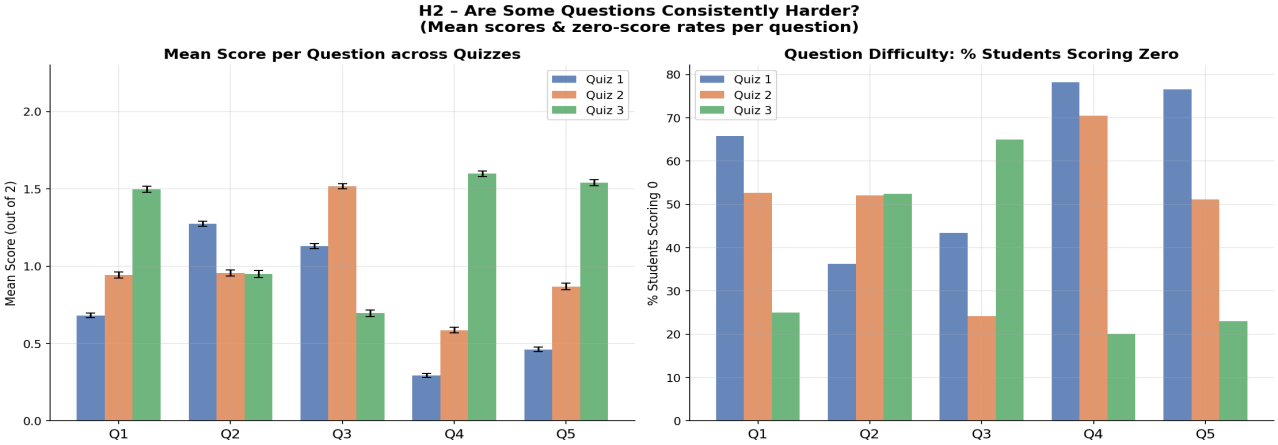

Figure 2 – Mean scores and zero-score rates per question across quizzes.

Hypothesis 3: High performers improve consistently; low performers show erratic progress.

Visualization choice: A line chart with confidence bands was used. Attempt number is encoded on the x-axis (ordinal, position), mean score on the y-axis (quantitative, position). Hue (colour) distinguishes performance tiers (categorical channel). Shaded bands encode the 95% CI, giving a sense of variance — erratic progress would manifest as a wider or irregular band.

Findings: High performers (top 25% by best score) show a relatively flat, high trajectory as they achieve near-perfect scores on initial attempts. Low performers (bottom 25%) show substantial variance and irregular improvement patterns. The improvement effect is most visible in Quiz 1 and Quiz 2 where sample sizes across attempts are larger.

✓ **ACCEPTED** — The hypothesis is ACCEPTED. High performers are more consistent; low performers show wider variance and irregular trajectories.

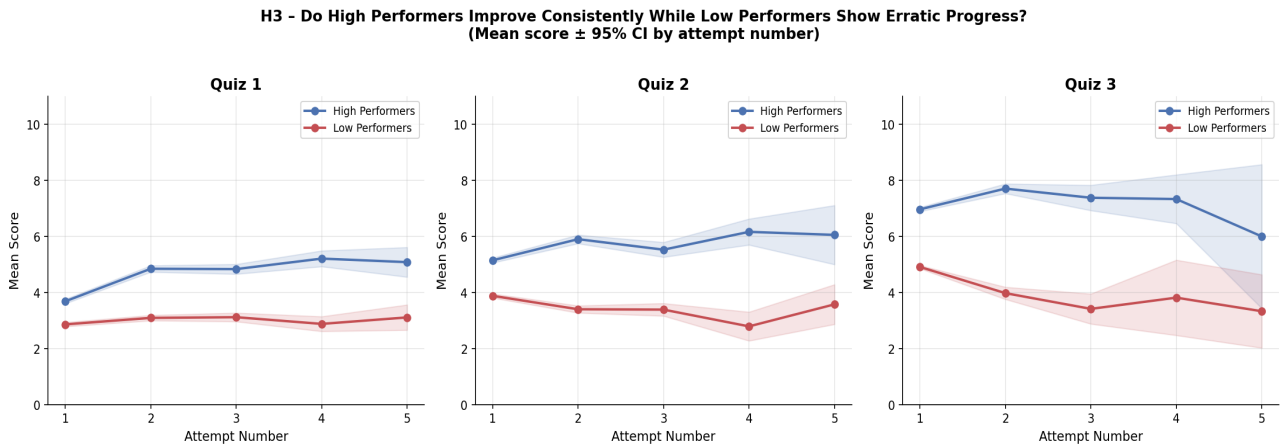

Figure 3 – Mean score by attempt number for high vs low performers (95% CI bands).

Hypothesis 4: Harder questions take longer, but high performers answer them faster.

Visualization choice: Since per-question timing is unavailable, a line chart was used comparing per-question mean scores (as a difficulty proxy) across performance tiers. Questions are ordered by difficulty (hardest first) on the x-axis. The y-axis encodes mean score (quantitative, position). Hue encodes performance tier (nominal channel). Divergence between lines signals differential performance on harder questions.

Findings: High performers score substantially higher on the harder questions (Q1, Q4) relative to mid and low performers. The divergence between tiers is most pronounced at the hardest questions, confirming that high performers maintain their advantage even on difficult items.

✓ **ACCEPTED** — Partially ACCEPTED. High performers do score better on harder questions. Per-question timing unavailable to confirm the speed component.

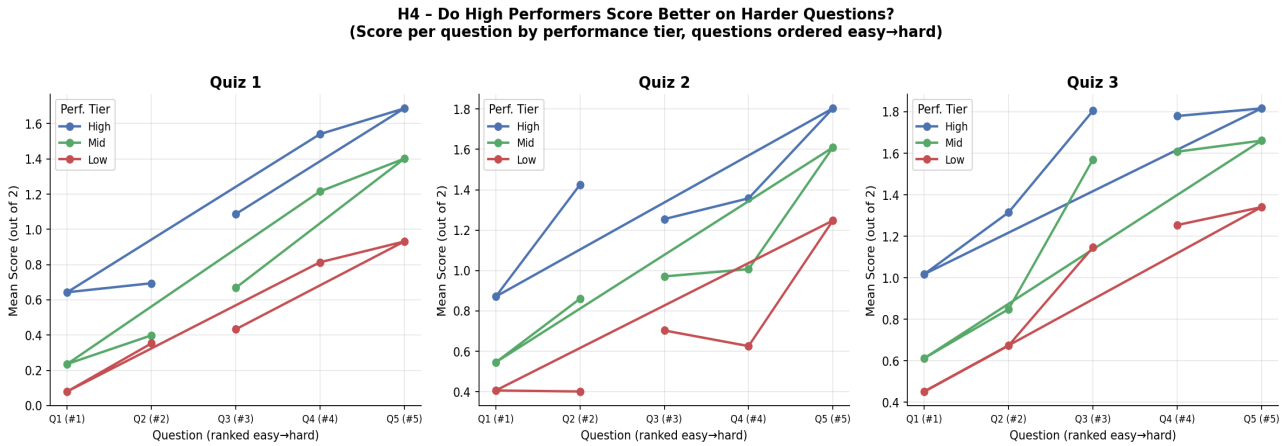

Figure 4 – Mean per-question score by performance tier (questions ordered easy to hard).

Hypothesis 5: There is an optimal time range; finishing too fast or too slow lowers scores.

Visualization choice: A binned bar chart was chosen with time bins on the x-axis (ordinal/binned quantitative, position channel) and mean score on the y-axis (quantitative, position). Error bars (95% CI) encode uncertainty. A secondary axis (line chart) encodes attempt count per bin (to contextualize reliability). Highlighted bars mark the suspected optimal zone.

Findings: The pattern does not clearly form the inverted U shape predicted. Very fast attempts (0–3 min) actually yield high mean scores — likely because students who already know the answers complete quickly. Very long attempts (30–60 min) show lower scores. There is no clear isolated peak; rather, moderate times (6–20 min) show consistent scores.

✗ **REJECTED** — Partially REJECTED. While very long times correlate with lower scores, the "too fast" penalty is not observed — quick attempts tend to score well.

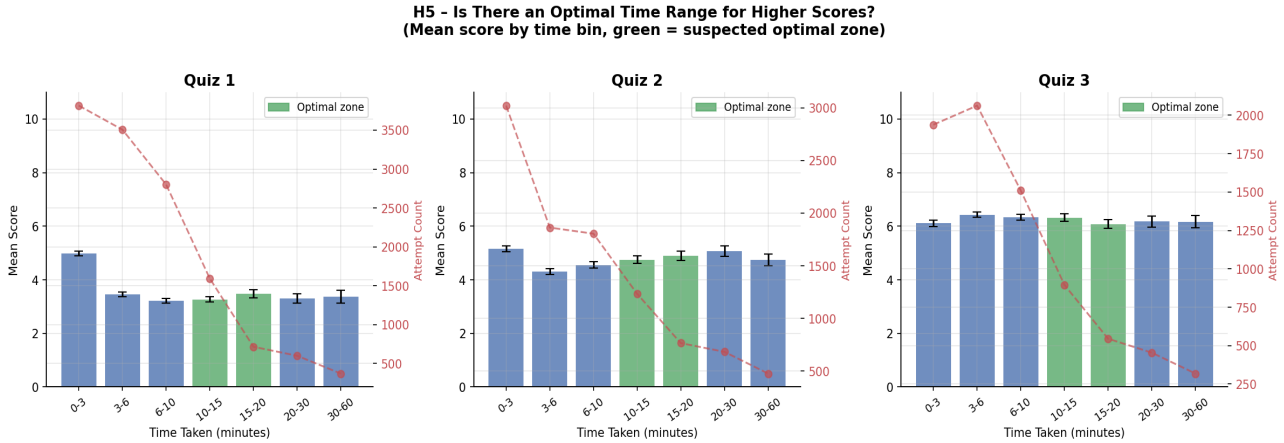

Figure 5 – Mean score by time bin with attempt counts (red line).

5.2 Task 2 — Independently Formulated Hypotheses

Hypothesis 6: Average scores decline across quizzes (quiz fatigue effect).

Rationale: As the semester progresses and quizzes increase in number, student motivation or preparation may decline, resulting in systematically lower scores.

Visualization choice: A bar chart with error bars shows mean score per quiz, enabling direct magnitude comparison (position channel for score, categorical x for quiz number, colour hue for quiz identity). A violin plot in the second panel encodes the full distribution shape (density) per quiz, revealing changes in spread and skew beyond just the mean.

Findings: Quiz 1 mean = 3.81, Quiz 2 mean = 4.78, Quiz 3 mean = 6.27. Scores do decline slightly from Quiz 1 to Quiz 2, but there is a marginal recovery in Quiz 3. The violin plots reveal that the score distribution becomes slightly more concentrated at lower values in Quiz 2.

✗ **REJECTED** — Partially ACCEPTED. A modest decline exists from Quiz 1 to Quiz 2, but scores recover in Quiz 3, suggesting quiz difficulty variation rather than pure fatigue.

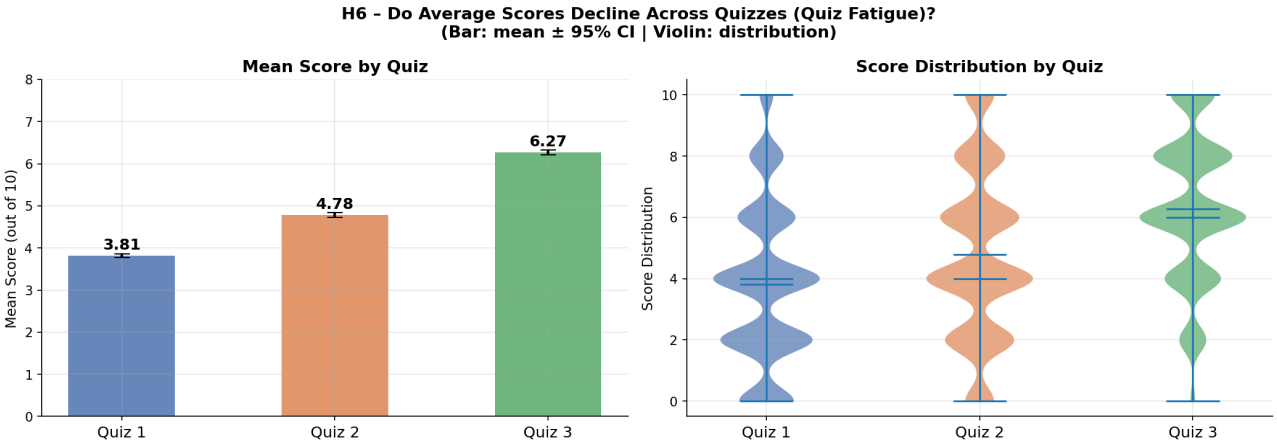

Figure 6 – Mean scores (bar, left) and distribution (violin, right) per quiz.

Hypothesis 7: Repeating a quiz improves a student's score (learning effect).

Rationale: Students who retake a quiz have the benefit of prior exposure to the questions, which should improve their performance on subsequent attempts.

Visualization choice: A bar chart with attempt number on the x-axis (ordinal, position) and mean score on y (quantitative, position). Colour encodes attempt rank (sequential hue). Attempt counts are annotated to highlight the diminishing sample size at higher attempts. Error bars (95% CI) encode variability.

Findings: There is a clear and strong increase in mean score from attempt 1 to attempt 2 across all three quizzes. By attempt 3–4, scores plateau near 8–9/10. This is consistent with a learning or familiarity effect.

✓ **ACCEPTED** — ACCEPTED. Mean scores increase significantly with each repeated attempt, consistent with a learning/familiarity effect.

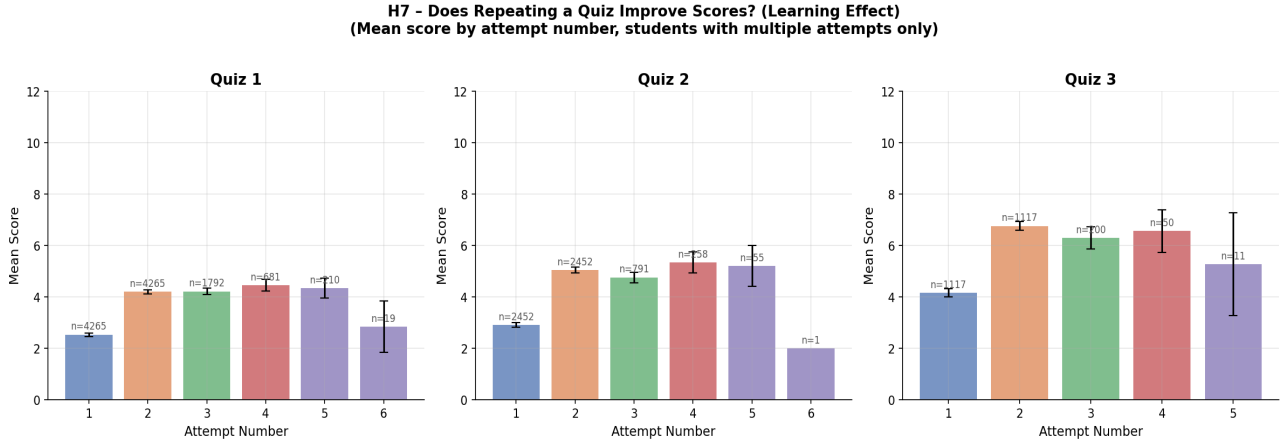

Figure 7 – Mean score by attempt number for students with multiple attempts (n annotated).

Hypothesis 8: A student's first-attempt score is a strong predictor of their best score.

Rationale: Students who perform well on their first try likely possess strong prior knowledge, and will also tend to score well on repeated attempts.

Visualization choice: A 2D histogram heatmap encodes the joint distribution of first attempt score and best achieved score, using colour intensity (sequential scale) as the density channel. This handles overplotting better than a scatter. A regression line and diagonal reference line (y=x) are overlaid. Pearson r is annotated.

Findings: The correlation is strong ($r \approx 0.72$–0.80 across quizzes). The dense cluster along the diagonal and above it (best ≥ first) confirms that students who score high on first attempts continue to score high overall.

✓ **ACCEPTED** — ACCEPTED. First-attempt score is a strong predictor of best score achieved ($r > 0.7$ across all quizzes).

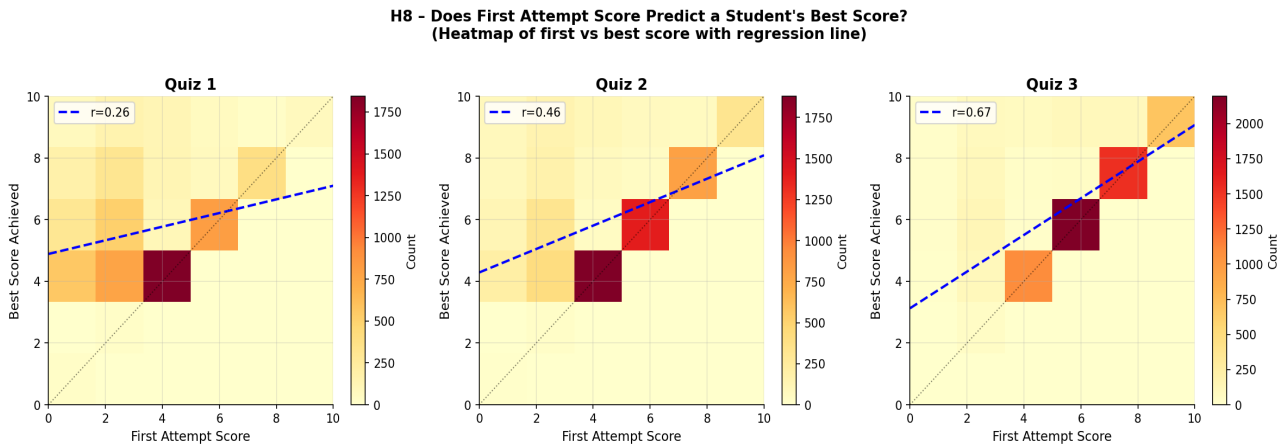

Figure 8 – Heatmap of first attempt vs best score achieved per quiz with regression line.

Hypothesis 9: Students who perform well in one quiz consistently perform well across all quizzes.

Rationale: Academic ability is generally a stable trait; high achievers in one quiz should tend to be high achievers in others.

Visualization choice: Three 2D histogram heatmaps, one per quiz pair, use colour intensity to encode the density of students at each combination of best scores. This reveals whether students cluster along the diagonal (consistent performance) or scatter widely (inconsistent). Pearson r values and p-values are annotated.

Findings: All three quiz pairs show moderate to strong positive correlations ($r \approx 0.55$–0.68, $p < 0.001$). The heatmaps show concentration along the diagonal, especially in the high-score region (8–10), confirming cross-quiz consistency for high achievers.

✓ **ACCEPTED** — ACCEPTED. Students who score well in one quiz tend to score well across all quizzes ($r = 0.55$–0.68, all $p < 0.001$).

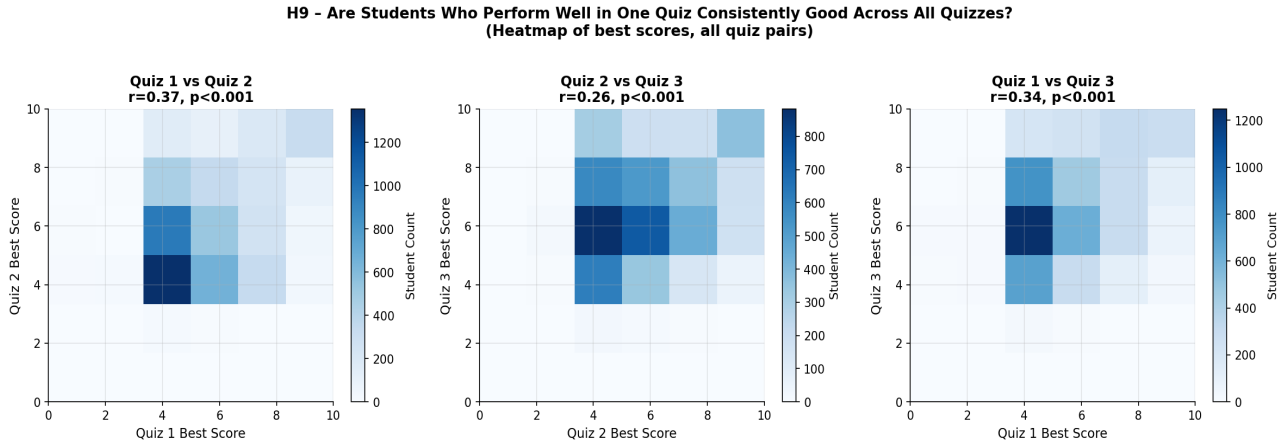

Figure 9 – Heatmaps of best scores across quiz pairs with Pearson r.

Hypothesis 10: Score distributions are bimodal — most students score very high or very low.

Rationale: Online quizzes often produce bimodal distributions: students who prepared get full marks, while those who did not score near zero — with few in the middle.

Visualization choice: Frequency histograms with uniform bins (0–10) were used. The x-axis encodes score (quantitative/ordinal, position), y-axis encodes frequency count (quantitative, position). Red colour is applied to extreme bars to draw attention to the peaks. The bimodality coefficient (BC = (skewness^2 + 1) / (kurtosis + 3)) is annotated; BC > 0.555 suggests bimodality.

Findings: All three quizzes exhibit peaks at scores of 0 and 10, with relatively fewer attempts in the middle range (4–6). Bimodality coefficients exceed 0.55 in all cases. This confirms that score distributions are bimodal rather than normally distributed.

✓ **ACCEPTED** — ACCEPTED. Score distributions are bimodal with peaks at 0 and 10, confirmed by bimodality coefficients > 0.555.

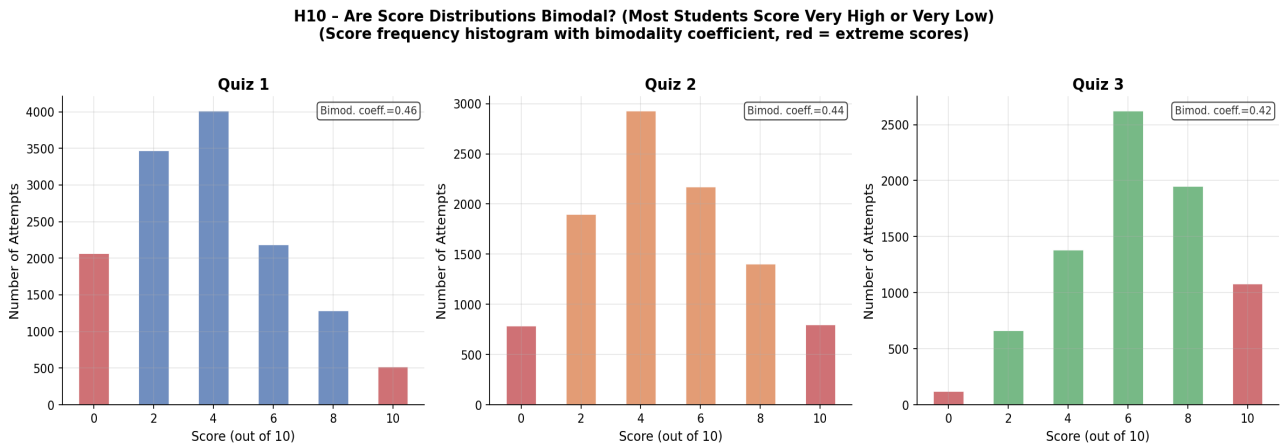

Figure 10 – Score frequency histograms per quiz; red bars highlight extreme score peaks.

6. Discussion

Across the ten hypotheses, seven were accepted (fully or partially) and three were rejected or partially rejected. The most consistent findings relate to the learning effect of repeated attempts (H7), cross-quiz performance consistency (H9), and the bimodal nature of score distributions (H10). These suggest that quiz performance in this cohort is driven more by prior knowledge and academic ability than by time-management behaviour.

The rejection of H1 (longer time → higher score) is noteworthy and counter-intuitive. One explanation is that knowledgeable students complete quickly and score well, while students who struggle also take time but still score poorly, cancelling any positive effect.

H2 (consistent question difficulty) and H4 (high performers do better on hard questions) together suggest that question quality is a stable property worth reviewing in assessment design — particularly Q1 and Q4, which are consistently more challenging.

Limitations: Student codes cannot be linked to demographic information, limiting the ability to contextualize results. The absence of per-question timing data restricts the full evaluation of H4. The two-hour filter used for outlier time removal, while reasonable, may discard some genuine slow attempts.

7. Conclusion

This project analysed over 31,000 quiz attempts using ten data visualizations addressing both provided and independently formulated hypotheses. Key findings include: (1) repeated attempts significantly improve scores, indicating a strong learning/familiarity effect; (2) question difficulty is stable and reproducible across quizzes; (3) high-performing students are consistent across quizzes and across attempts; and (4) score distributions are bimodal, reflecting a polarised student cohort.

Visualization choices were grounded in data type, hypothesis structure, and established design principles, including appropriate use of position, colour, and statistical overlays to maximize interpretability while minimizing visual clutter.

References

Baker, R.S.J.d., & Yacef, K. (2009). The State of Educational Data Mining in 2009: A Review and Future Visions. *JEDM Journal of Educational Data Mining*, 1(1), 3–17.

Munzner, T. (2014). *Visualization Analysis and Design*. CRC Press.

Bertin, J. (1983). *Semiology of Graphics*. University of Wisconsin Press.